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Failure Prediction of Composite Open Hole Tensile Test Specimens Using Bond Based Peridynamic Theory

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Abstract

In this study, interaction of holes in different locations in a tensile loaded composite plate is analysed using Bond-Based Peridynamic (PD) Theory. Moreover, the crack growth behaviour in the plate is investigated. PD simulations are conducted using adaptive dynamic relaxation solution scheme using a code developed in MATLAB. The PD solutions of the crack tip opening displacement is validated with the results obtained through finite element analysis (FEA) using ANSYS. The crack propagation paths and the effect of hole interaction on crack growth behaviour are also investigated using PD theory. Results obtained using PD theory were compared with the results of FEA and Özasan et al. (2019). It is found that PD can accurately capture stress, strain and hole interactions in composite laminates.

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Nomenclature

CFRP	Carbon Fiber-Reinforced Plastic	NOHT	Unnotched Specimen
FEA	Finite Element Analysis	OHT	Open Hole Tension
GFRP	Glass Fiber-Reinforced Plastic	PD	Peridynamics

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1. Introduction

Composite materials are preferred due to their lightness and strength selection in the required direction. In orthotropic materials, the number and interaction of failure modes are high, so damage modelling is important and difficult. For this reason, it is crucial to perform damage analysis correctly. Damage analysis is even more essential, especially if there is a hole that creates stress concentration in the structure. The complex damage mechanism of composite laminates with holes have increased the interest of researchers in this subject. The effects of geometrical parameters on the failure modes of laminated composites were investigated by Karakuzu et al. (2008). A woven-glass-vinylester plate with two serial pin loaded hole was analyzed experimentally and numerically. The purpose of this study was to investigate the effects of geometrical parameters such as edge distance, plate width and distance between two holes to hole diameter ratios on failure loads and failure modes. By increasing all three ratios the ultimate load capacity was increased. Then stress distribution between two holes in carbon fiber laminates was analyzed by Ghezzi et al. (2008). Holes were set orthogonal and parallel with respect to load direction for different symmetric carbon fiber laminates. Stress distribution was investigated under in-plane tension loading. For the specimens with holes located parallel to the load direction, they concluded that the stress concentration factor is not affected by the distance between holes. It is also stated that hole interaction begins when the distance between holes located orthogonally to the load direction was set to 1.5 times the hole diameter. Özaslan et al. (2019) studied interaction between two holes specified in five different angle orientation for carbon/epoxy laminates. The study was performed by experimental and numerical methods. As a result, it was found that the location of stress concentration depends highly on the hole orientations. It was stated that the damage around the holes would initiate at the high stress concentration areas. By O'Higgins et al. (2008) an experimental study was carried out of a carbon fiber-reinforced plastic (CFRP) and glass fiber-reinforced plastic (GFRP) laminates to determine the open hole tension (OHT) characteristics. Damage progression was mapped during the tension test where ultimate-load were applied. The damage progression mechanisms were determined to be similar for both materials. However, the CFRP specimens were found to be stronger, while the GFRP specimens were found to have greater ultimate strain.

Peridynamic(PD) theory is a new and developing theory. PD is a nonlocal theory present by Silling (2000) in 2000. That a material point is effected by the material points inside a finite radius is called nonlocal theory. The PD theory is basically a reformulation of the equation of motion in solid mechanics. Mathematically the theory uses spatial integral equations, which make it easy to model discontinuities. The introduced PD theory in 2000 have been later named "Bond Based" PD theory. As mentioned above PD theory has an advantage in modelling discontinuities, Silling (2003) considered Kalthoff-Winkler experiment on a plate with two parallel notches. The PD theory ability to observe crack prediction and its angle were found. To solve dynamic problems using PD theory a numerical method was introduced by Silling and Askari (2005). This study demonstrates the characteristics of the method of modelling brittle dynamic crack growth. The biggest advantage of PD theory to fracture mechanics is that it does not require any complementary relationship that specifies when the crack should branch, stop, and orient randomly. For orthotropic materials Oterkus and Madenci (2014) have present an application of PD theory to predict how damage propagates in fiber-reinforced composite materials under mechanical and thermal loads. Problems have been analyzed in fiber direction and arbitrary directions. Therefore, two PD bond constants have been determined for each direction. Failure mechanisms as matrix crack, fiber breakage and delamination were examined. They came to the conclusion that PD theory is successful for failure analysis of composite materials. Investigation of damage mechanism of composite laminates subjected to low-velocity impact was performed by Askari et al.(2006) using PD theory. Later did Xu et al.(2008) and Colavito et al.(2007) research under the same problem and the results showed that PD simulations is quite successful. Notched composite laminates in addition was analysed by Xu et al.(2007) under biaxial loads. Then Kilic et al. (2009), did a research on PD theory using two-ply and three-ply laminates with different laminae orientations with a center crack. The PD results was validated with experimental data. The failure mechanisms; matrix cracking, fiber breakage, delamination were captured by the prediction. The reported experimental results showed the ability of the PD theory to realistically capture failure initiation and propagation in composites.

In this paper, carbon/epoxy laminates with open holes are investigated under uniaxial tension. Bond based PD theory was performed in order to understand the interaction mechanisms of holes located in different positions.

2. Problem Description

In this study, the solution of the problem described in Fig. 1. where composite plates are subjected to a uniaxial static tensile load is obtained using PD theory. First, PD solution of stress and strain curve for the unnotched specimen (NOHT) in Fig. 1. is obtained and the results are validated with experimental data in Öztaşlan et al. (2019). In order to understand the interaction of holes, stresses and displacements along the paths shown in Fig. 1. are calculated for the three different carbon/epoxy composite specimens. The first specimen, OHT1 is a composite plate with a single central hole. The other two specimens are labelled OHT2-1 and OHT2-2 and have two holes located orthogonally to load direction with a distance of $P = 3D$ and $P = 1.5D$, between the holes respectively. Note that the diameters of the holes are the same for all specimens. The geometrical properties are provided in Table 1.

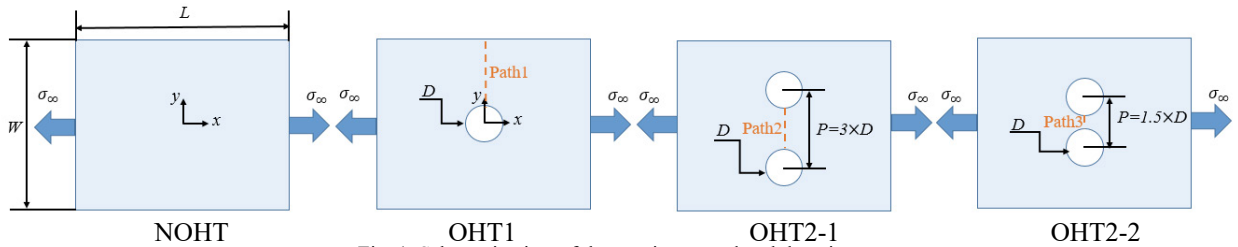


Fig. 1. Schematic view of the specimens and path locations.

Table 1. Geometrical Dimensions of the Specimens.

Specimens	NOHT	OHT1	OHT2-1	OHT2-2
Width, W	36 mm	36 mm	36 mm	36 mm
Length, L	250 mm	250 mm	250 mm	250 mm
Thickness, t	2.24 mm	2.24 mm	2.24 mm	2.24 mm
Hole diameter, D	-	4 mm	2×4 mm	2×4 mm
Path Length, P	-	16 mm	12 mm	6 mm

The composite specimens shown in Fig. 1. have 8 layers of SIGRAPREG C U310-0/NFE340/ 35% unidirectional epoxy carbon fiber prepreg with the layup of $[45/90/-45/0]_s$. The elastic constants of a single ply are given in Table 2. The applied load on the two edges of the specimens are $\sigma_\infty = 700$ MPa for the unnotched specimen and $\sigma_\infty = 372.02$ MPa for the open hole specimens.

Table 2. Lamina Material Properties.

Material Properties	
E_1	130 GPa
E_2	9.4 GPa
G_{12}	3.5 GPa
ν_{12}	0.33

First, the bond based PD models of the specimens in Fig. 1. are generated and solved with an in-house script developed in MATLAB R2015a (2015). PD results are obtained in terms of displacement field from which the strains and stress are derived. Then, finite element analysis (FEA) are performed for the same cases using ANSYS (2013). The distribution of the PD displacements, strains and stresses are compared with the FEA results and experimental data from Öztaşlan et al. (2019).

3. Methods

In this paper, two different approaches are used to solve the aforementioned problem in Section 2. These are PD theory and FEA. Directional displacements, strain, normal stress and crack propagation simulations are modelled by

the PD theory. Normal strains, normal stresses and directional displacements are obtained using FEA to validate PD results.

3.1. Bond Based Peridynamic Theory

Bond based PD theory is a nonlocal theory of continuum mechanics with displacement formulation Silling (2000). PD approach is valid at both continuous structures and discontinuities since the equation of motion has an integral formulation as

$$\rho(x)\ddot{\mathbf{u}}(x,t) = \int_H \mathbf{f}(x' - x, u' - u) dH + \mathbf{b}(x,t), \quad (1)$$

where $\rho(x)$ is the mass density, $\ddot{\mathbf{u}}(x,t)$ is the acceleration at material point x , $\mathbf{f}$ and $\mathbf{b}(x,t)$ denote PD force density and body force density, respectively. In the PD theory, a material point interacts with the other material points inside a surrounding region with a finite radius, δ called horizon. In the current study, the size of the horizon is taken as $\delta = 3.015\Delta x$ as suggested in Oterkus et al. (2010) where Δx is the spacing between material points. For a material point x , material point family H_x is composed of all the material points inside the circle with a horizon radius, δ . The PD force density, f between the material points x and x' can be calculated as in Eq.(2)

$$f(x' - x, u' - u, t) = 2\delta(\mu_F c_f + c_m) s \frac{y' - y}{|y' - y|}, \quad (2)$$

where $x' - x$ is the relative position vector and $u' - u$ is the relative displacement vector. The position of the deformed material points x and x' is expressed as y and y' respectively. The term s in Eq. (2) is the stretch between material points that can be calculated as

$$s = \frac{|y' - y| - |x' - x|}{|x' - x|}. \quad (3)$$

Two bond constants are required for orthotropic materials since their properties are directional dependent Oterkus et al.(2012). The bond constant that is present in the fiber direction is c_f and the other bond constant c_m is present in all directions. The bond direction is required for the fiber bond constant, when the bond is parallel to the fiber direction μ_F become 1 as given in Eq. (4). c_f and c_m can be calculated in terms of engineering constants as follows

$$\mu_F = \begin{cases} 1 & (x - x') // \text{fiber direction} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

$$c_f = \frac{Q_{11} - Q_{22}}{\frac{1}{2} \sum_{j=1}^N \xi_{ji} V_j}, \quad c_m = \frac{8Q_{22}}{\pi t \delta^3}, \quad (5)$$

where ξ is the initial distance between the material points and V is the volume of material point. Q_{11} and Q_{22} in Eq. (5) are stiffness matrix components and they can be calculated as

$$Q_{11} = \frac{E_{11}}{1 - \nu_{12}\nu_{21}}, \quad Q_{22} = \frac{E_{22}}{1 - \nu_{12}\nu_{21}} \quad (6)$$

The bond constants in Eq. (5) are derived for a single ply. Laminates with multiple layers in Özaslan et al. (2019) are considered in this study. Therefore, the material constants of the laminate are homogenized by calculating equivalent stiffness properties in fiber and transverse directions. First, extensional compliance matrix A^* and extensional stiffness matrix A of the laminate are calculated as Kaw (2005)

$$A_{ij} = \sum_{k=1}^n \left[(\bar{Q}_{ij}) \right]_k (h_k - h_{k-1}), \quad i = 1, 2, 6; j = 1, 2, 6, \quad (7)$$

$$[A^*] = [A]^{-1} \quad (8)$$

where h_k is the thickness of ply k and $\bar{Q}_{ij}$ are the elements of the transformed reduced stiffness matrix from Kaw

(2005). The elastic modulus in loading direction E_x , normal to loading direction E_y , in-plane shear modulus G_{xy} and the Poisson's ratio ν_{xy} of the laminate then can be calculated as

$$E_x = \frac{1}{hA_{11}^*}, \quad E_y = \frac{1}{hA_{22}^*}, \quad G_{xy} = \frac{1}{hA_{66}^*}, \quad \nu_{xy} = -\frac{A_{12}^*}{A_{11}^*}. \quad (9)$$

The equivalent engineering constants in Eq. (9) are substituted in Eq. (6) in order to calculate the bond constants c_f and c_m .

In PD, material failure can be considered by breaking the bonds between material points when the stretch in Eq. (3) reaches a critical value s_0 . The critical stretch, s_0 can be related to the fracture toughness G_0 by calculating the PD energy to break all the bonds per unit area Ha et al. (2010).

$$G_0 = \frac{9s_0^2 E \delta}{4\pi}. \quad (10)$$

PD mesh of the composite specimens with open holes are shown in Fig. 2a. Zoomed view of the PD mesh and the material points are illustrated in Fig. 2b. In this study, the grid size in OHT1 and OHT2-1 specimens are $\Delta x = 0.667$ mm. In OHT2-2, the grid size is reduced to $\Delta x = 0.5$ mm or more accuracy, since path3 is the shortest path.

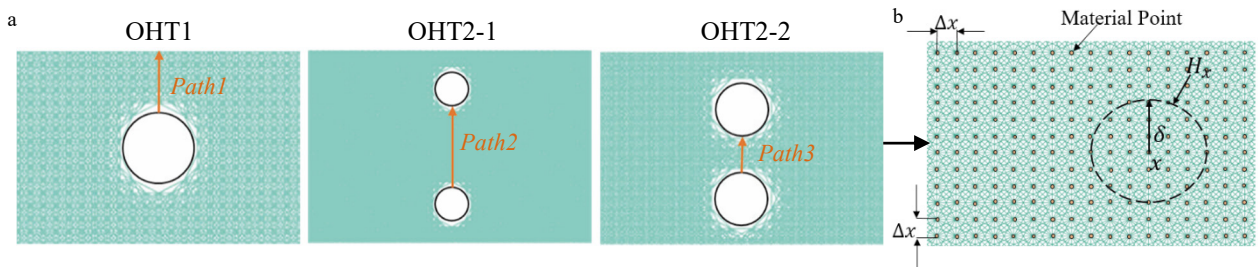


Fig. 2. Peridynamic mesh a) general view with holes b) zoomed view.

3.2. Finite Element Method

FEA solution of the problem defined in Section 2 is conducted in ANSYS software. The FEA model of the specimens are generated with shell181 elements as shown in Fig. 3. The element size around the holes for OHT1 is $\pi/18 = 0.175$ mm, OHT2-1 and OHT2-2 are $\pi/20 = 0.157$ mm. To design a fine mesh region as shown in Fig. 3. specimens are divided on the symmetry planes, squares with the holes in the center are defined and rings are defined to get annular mesh around the holes as in Fig. 3. The equivalent material properties of the laminate in Özaslan et al. (2019) are calculated using Eq. (9).

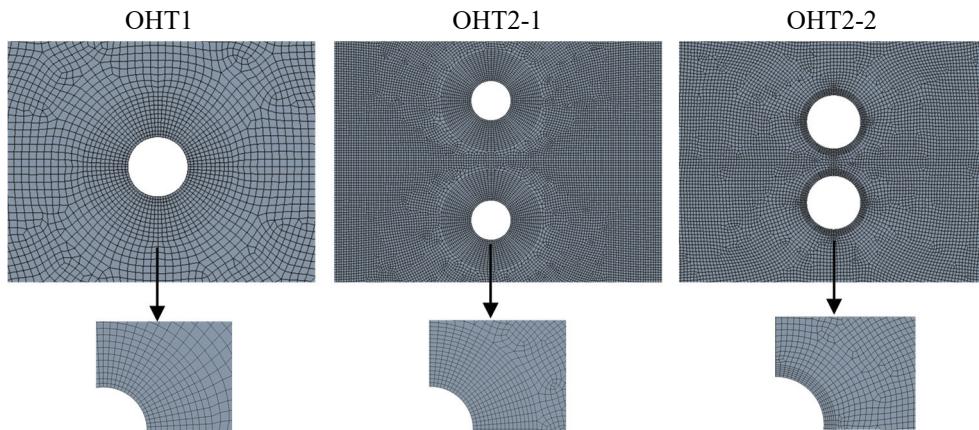


Fig. 3. FEA mesh around the holes of the specimens

4. Results

In this study, static solution of composite laminates subjected to uniaxial tensile load are performed using PD theory. The geometrical and material properties of the specimen are explained in Section 2. First, linear elastic PD solution of the unnotched composite specimen is performed under uniaxial tensile stress of $\sigma_{\infty} = 700$ MPa. The PD results of the stress-strain behavior of the unnotched specimen is obtained as in Fig. 4. The PD solution is compared with the experimental results in Özaslan et al. (2019) in Fig. 4. Note that, NOHT1, NOHT2 and NOHT3 denote the three different test specimens that have the same geometrical and material properties with the PD solution. The comparison of PD results and experimental data in Fig. 4. indicate that PD model is able to capture the linear elastic behavior of the unnotched composite specimen.

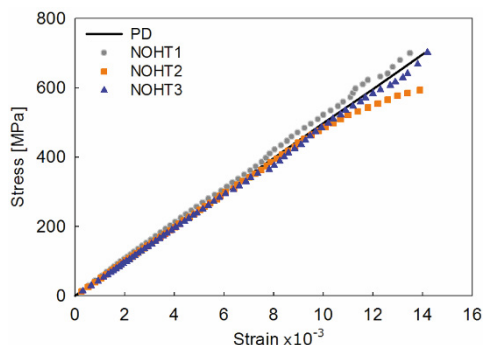


Fig. 4. Comparison of PD strain values with experimental data from Özaslan et al. (2019).

Next, PD and FEA solutions of the three different plates with holes subjected to an axial load of $\sigma_{\infty} = 372.02$ MPa are obtained. The distributions of normal stresses obtained using PD and FEA are shown in Fig. 5. The PD stresses are derived from the strain field which is calculated using the displacements of two consecutive points. For both PD and FEA, the stress distribution contours around the holes are similar qualitatively. However, the PD solutions yield lower stress values around the holes. Far field stresses is $\sigma_{\infty} = 372.02$ MPa for both FEA and PD solutions. It is observed that, the maximum normal stress of the plate with single hole (OHT-1) is lower than the specimens with double holes (OHT2-1 and OHT2-2). Both maximum PD and FEA normal stresses for the OHT2-1 specimen is very close to OHT-1 specimen which indicates that, there is no stress interaction of the two holes for the OHT2-1 specimen. Note that, the distance between the holes is $P = 3D$ for the OHT2-1 specimen. As the distance between the holes is reduced to $P = 1.5D$, (OHT2-2) the maximum stresses increase due to interaction between holes as seen in the normal stress distribution of the OHT2-2 specimen in Fig. 5.

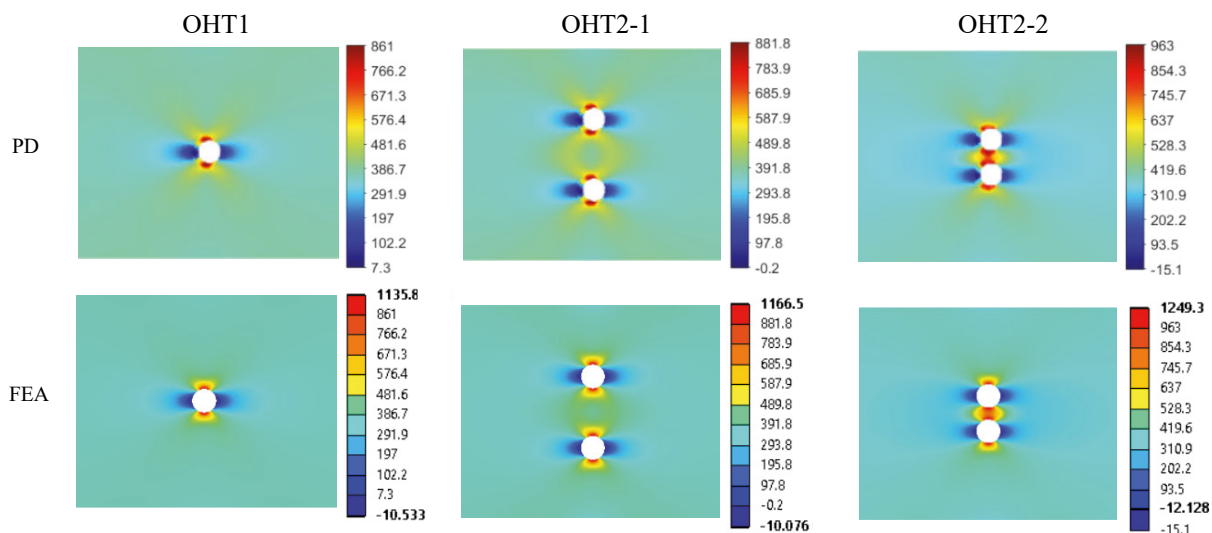


Fig. 5. Normal stress distributions for open hole tension specimens calculated with PD and FEA.

Next, the variation of normal stresses along the paths shown in Fig. 1. are calculated. PD results are compared with FEA and results in Özaslan et al. (2019) in Fig. 6. It is seen that, the PD results are identical with FEA and the results in Özaslan et al. (2019) for OHT-1 specimen. The PD results in agreement with FEA and Özaslan et al. (2019) for OHT2-1. FEA results are nearly overlaps Özaslan et al. (2019) for all specimens. The PD results of normal stresses between the holes deviate from FEA and Özaslan et al. (2019). for OHT2-1 and OHT2-2 specimens. However, the shape of the normal stress profiles for OHT2-1 and OHT2-2 specimens are in agreement with FEA and Özaslan et al. (2019).

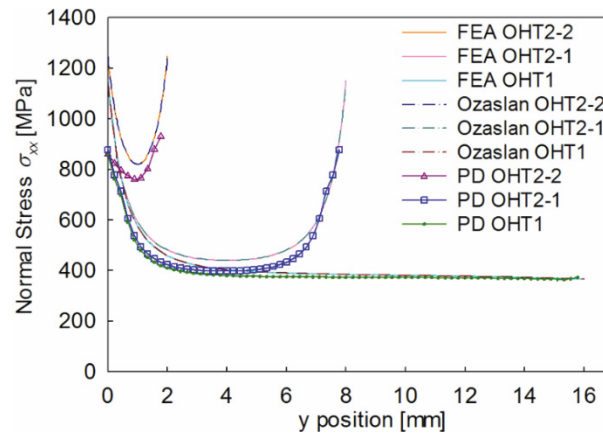


Fig. 6. Comparison of PD and FEA results of normal stress along the paths with Özaslan et al. (2019).

The variation of normal strains in the middle of the plates with respect to y coordinate are obtained using PD and FEA as shown in Fig. 7. The PD and FEA results are in very good agreement for OHT1 specimen. It is seen that, the FEA solution yields higher normal strain values around the hole for OHT1 specimen. The PD strains away from the holes are identical with FEA for OHT2-1 and OHT2-2 specimens and the discrepancy between PD and FEA increase around the holes.

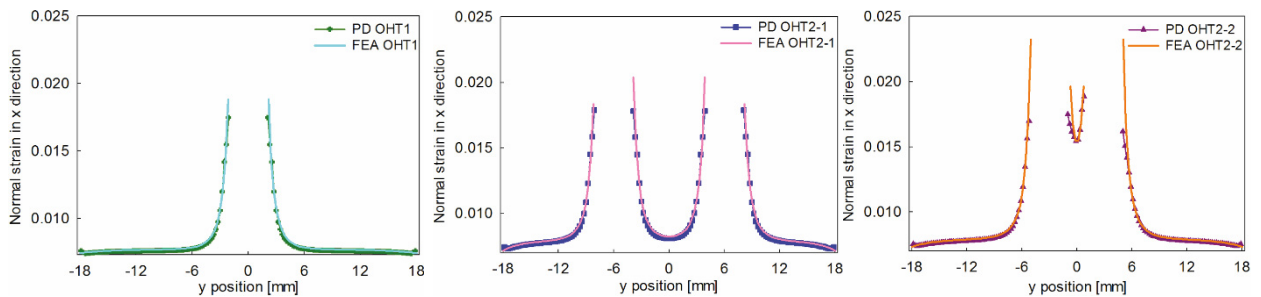


Fig. 7. PD and FEA results of normal strains in loading direction.

The PD displacements in y and x directions as a function of y and x coordinates on the lines at $x = 0$ and $y = 0$ respectively of the plate are compared with FEA in Figs. 8a. and 8b., respectively. The PD displacement profiles in Fig. 8. overlap FEA results. The comparison PD displacement contours in y and x directions are given in Fig. 9. and Fig. 10., respectively. PD displacement fringes of all specimens are in very good agreement with FEA results.

The initiation and propagation of cracks in the OHT1, OHT2-1 and OHT2-2 specimens subjected to uniaxial tensile load is calculated using PD. The evolution of the cracks is illustrated in Fig. 11. It is observed that, the cracks initiate around the holes where the normal stresses are maximum as seen Fig. 5. For the specimens with double holes (OHT2-1 and OHT2-2), the crack propagates between the holes and then propagates through the free edges.

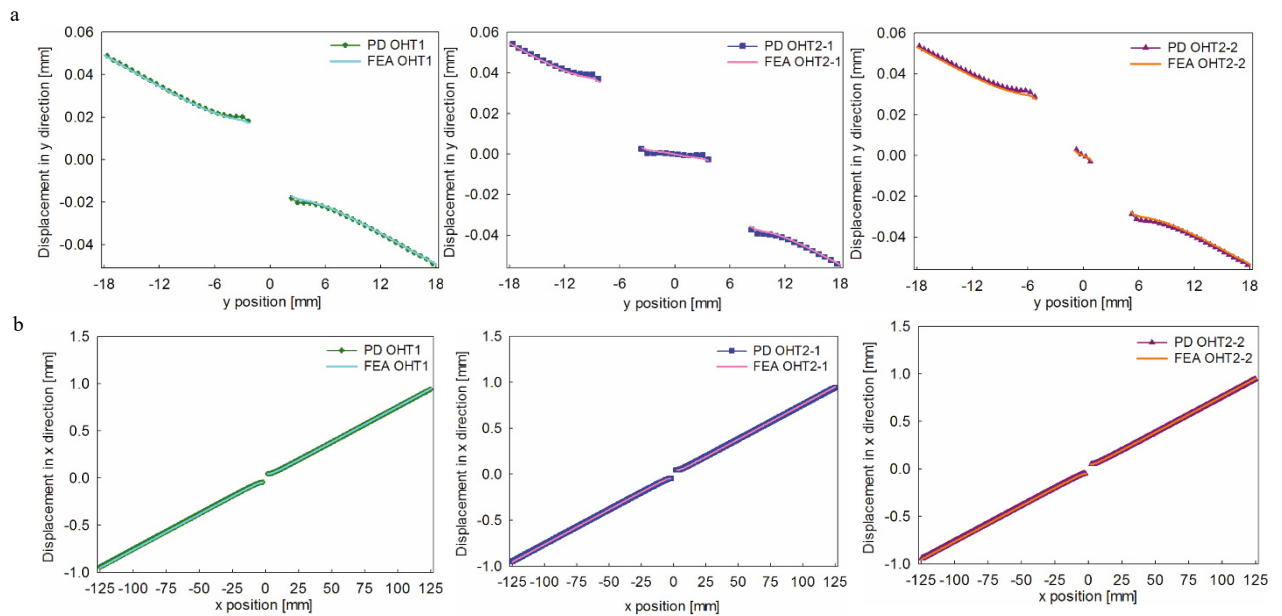


Fig. 8. PD theory displacement results validation with FEA in a) y direction and b) x direction.

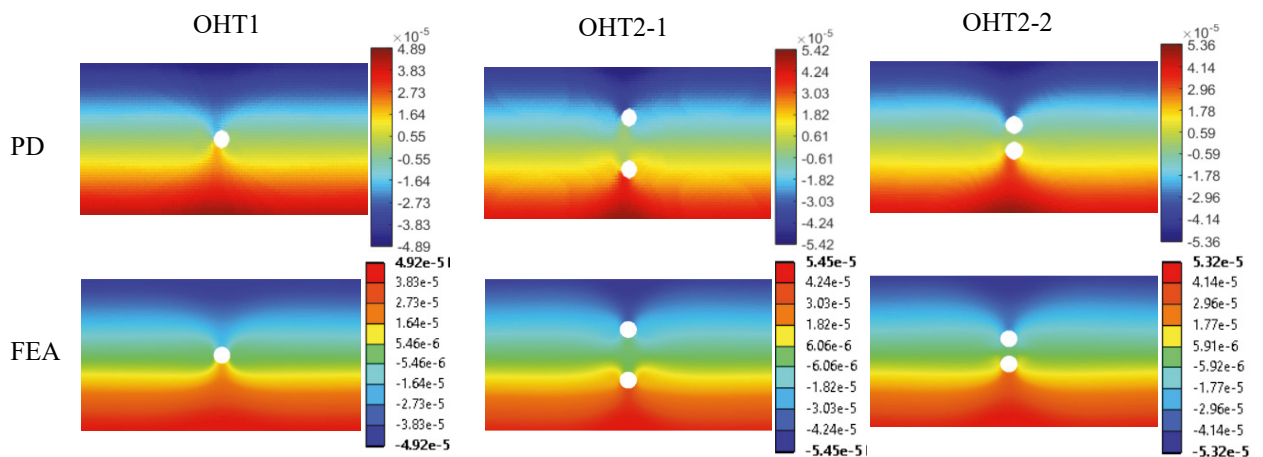


Fig. 9. Displacement contours in y direction.

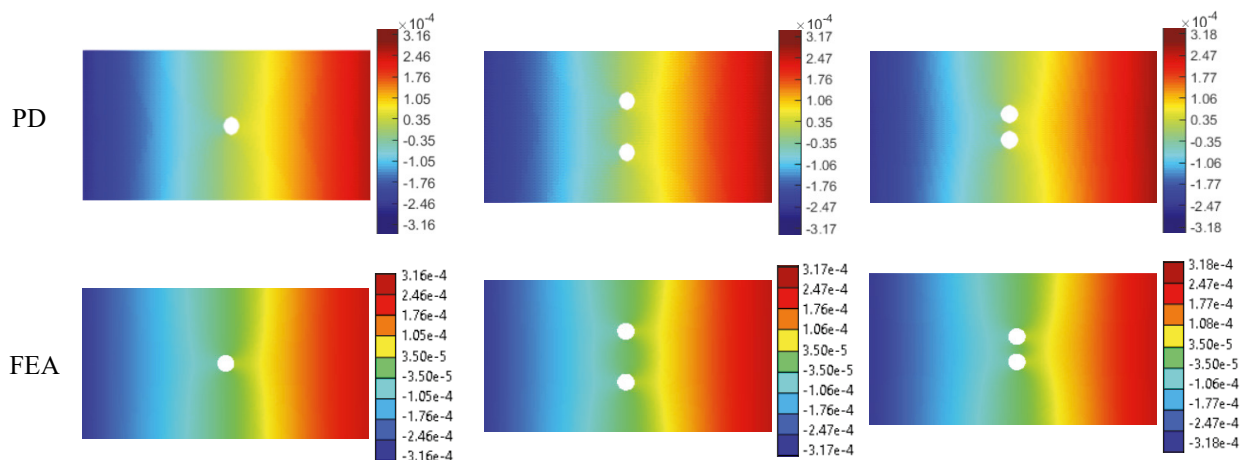


Fig. 10. Displacement contours in x direction.

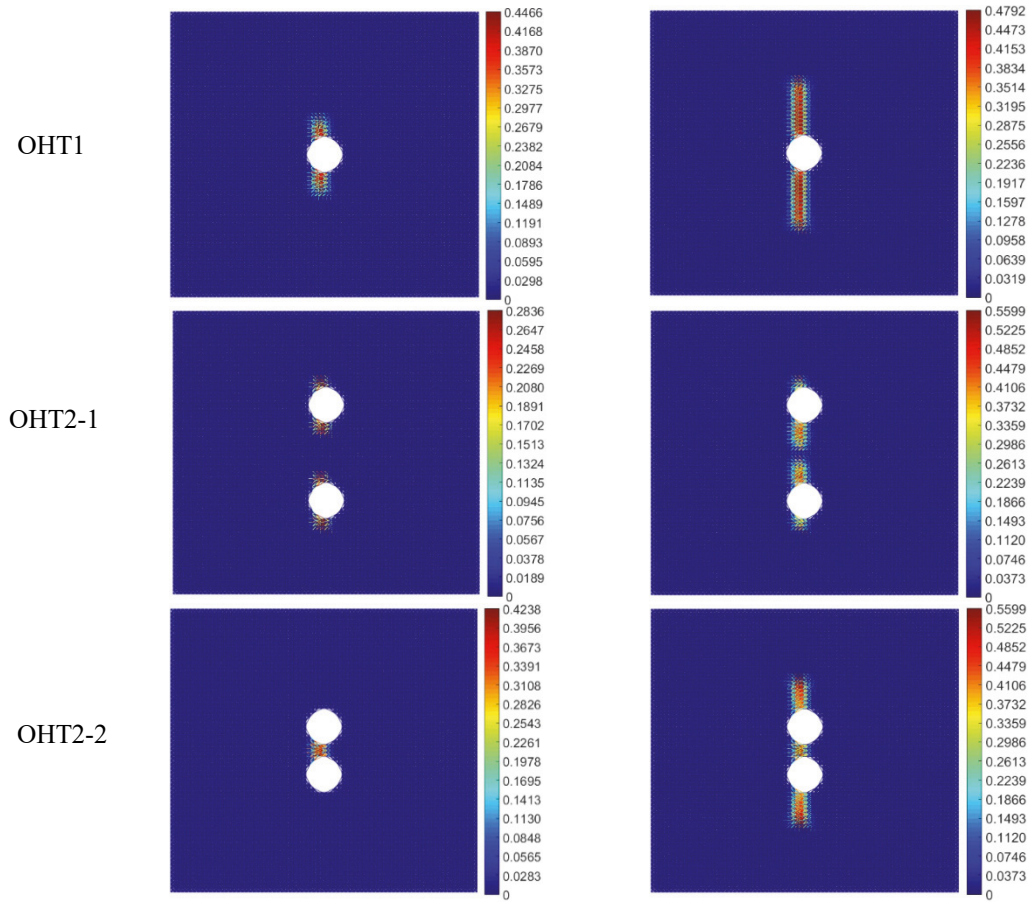


Fig. 11. Crack propagation.

5. Conclusion

This open hole interaction study of carbon/epoxy laminates under tension is carried out with bond based PD theory. The distribution of the PD displacements, strains and stresses are compared with the FEA results and experimental data from Özaslan et al. (2019) and the results obtained are consistent. The PD model validation with the experimental data from Özaslan et al. (2019) of the unnotched composite specimen under tension, showed that PD model is able to capture the linear elastic behavior of the specimen. Then, the normal stress distributions are evaluated to compare the specimens. It is concluded due to stress distribution and maximum stress value that the holes in the OHT2-2 specimen with the distance of 1.5 times the diameter is interacting. The PD result of normal strains, gives identical values with FEA away from the holes. However, the FEA solution on the hole edges are higher than results obtained with PD theory. It should be mentioned that the PD stresses are derived from the strain field which is calculated using the displacements of two consecutive points. The PD displacement results in y and x directions of all specimens are in very good agreement with FEA results. Last, the modelling of the crack initiation and propagation with PD theory is compared to the stress distribution. It is observed that, the cracks propagate around the holes where the normal stresses are maximum. For the specimens with double holes the crack propagates between the holes and then propagates through the free edges.

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